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School of Computer Science and Engineering

DA1 REPORT

Problem Identification, Literature Survey, Problem Statement, and Proposed Architecture

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PRISM

Provenance & Reasoning Intelligence System for Multimedia Cross-Domain Multimodal Deepfake Detection and Forensic Analysis

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1. Problem Identification: Domain and Motivation

1.1 Application Domain

Digital media forensics and visual content authentication have emerged as essential pillars of cybersecurity, judicial evidence integrity, and public information resilience. The rapid democratization of advanced generative architectures, in particular latent diffusion models, generative adversarial networks (GANs), and neural vocoders, has enabled the synthesis of hyper-realistic digital artifacts at a scale that was previously inaccessible outside specialized research laboratories.

Modern media manipulation now extends well beyond simple full-face replacement to include fine-grained facial reenactment, synthetic voice cloning, background neural texture synthesis, and cross-modal audio-visual lip desynchronization. While such tools offer legitimate creative utility in filmmaking, dubbing, and accessibility, they simultaneously introduce serious vulnerabilities across corporate identity verification, legal digital-evidence chain-of-custody procedures (ISO/IEC 27037), financial fraud prevention, and the broader integrity of the information ecosystem.

1.2 Domain Quantification

The scale and technical complexity of synthetic media manipulation can be demonstrated through empirical metrics drawn from benchmark forensic datasets and security surveys:

- **Dataset volume and complexity:** Standard forensic benchmarks such as *FaceForensics++* contain over 1,000 original source videos and more than 500,000 manipulated frames, evaluated across raw, lightly compressed, and heavily compressed (H.264) tiers to reflect realistic distribution conditions.
- **Spatial localization challenge:** Over 70% of real-world deepfake attacks are localized, modifying specific facial sub-regions, mouth movements, or audio tracks while leaving the surrounding spatial and background geometry untouched. This makes whole-frame binary classification an incomplete forensic signal.
- **Cross-domain performance collapse:** Conventional binary classifiers experience a performance degradation of 20% to 35% in ROC-AUC (Receiver Operating Characteristic Area Under Curve) when evaluated on unseen generative models or media subjected to social-media lossy re-compression, commonly referred to as laundering.

1.3 Stakeholders and Decision Support

The proposed framework is designed to directly support three primary operational stakeholders:

1. Digital forensic investigators, who require verifiable, pixel-level spatial forgery masks and confidence reliability maps to differentiate localized edits from legitimate global camera adjustments.
2. Judicial and legal auditors, who require auditable, natural-language forensic summaries detailing detected structural, spectral, and temporal anomalies suitable for expert-witness testimony in legal proceedings.

3. Automated content-moderation platforms, which require calibrated frame-authenticity confidence scores to automate the filtering and flagging of synthetic media prior to widespread dissemination.

2. Literature Survey

To establish the current technological state of the art, fifteen representative research studies are evaluated below. In line with academic reporting guidelines, twelve studies are drawn from archival Q1/Q2 journals and indexed conference proceedings, with the remaining three restricted to leading computer-vision conference venues.

2.1 Comparative Literature Table

Ref.	Year	Venue & Type	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
[1]	2019	IEEE ICCV (Conference)	FaceForensics++ (FF++)	Fine-tuned XceptionNet backbone on frame crops	Accuracy ≈ 96–99% in-domain	Overfits to method-specific blending artifacts; fails on unseen generative engines
[2]	2021–22	IEEE T-CSVT (Journal)	FF++, DFDC	Sparse video transformer capturing spatiotemporal features	Cross-dataset AUC improvement over frame-level baselines	Video-level binary classification only; lacks spatial forgery masks and text reports
[3]	2023	IEEE/CVF CVPR (Conference)	FF++, WildDeepfake, CocoGlide	TruFor: dual-branch fusion combining RGB and Noiseprint++ residuals	F1 ≈ 0.98 on pristine regions; strong cross-manipulation generalization	High computational overhead; does not produce natural-language textual reports
[4]	2023	IEEE/CVF ICCV (Conference)	FF++, Celeb-DF v2	Thumbnail-layout video encoding for spatiotemporal inconsistency learning	State-of-the-art cross-manipulation AUC via composite frame grids	High GPU memory footprint due to spatial tiling and composite grid rasterization
[5]	2021	IEEE/CVF ICCV (Conference)	FF++, FFIW	Reference-assisted, identity-driven forgery detection	Improved robustness on identity-swap manipulations	Requires a confirmed authentic reference image of the target subject at inference
[6]	2020	ECCV (Conference)	FF++, Celeb-DF	Frequency-aware analysis via Discrete Cosine Transform (DCT) residuals	Consistent accuracy gains when fused with RGB-only baselines	Sensitive to heavy spatial blurring; degrades on non-JPEG container formats
[7]	2023	IEEE/CVF CVPR (Conference)	FF++, DFDC	Identity-leakage-aware modelling enforcing pristine decision bounds	Improved cross-dataset generalization by suppressing identity shortcut cues	Computational complexity scales poorly when extended to larger vision backbones
[8]	2020	Information Fusion (Journal Survey)	Multimodal benchmarks	Comprehensive survey on multimodal media authentication	Consolidated taxonomy across 100+ prior detection methods	Review paper; does not introduce a novel inference model
[9]	2022	IEEE ICASSP (Conference)	ASVspoof 2021	AASIST: heterogeneous graph attention network on raw audio waveforms	EER ≈ 0.83% on ASVspoof2019 LA benchmark	Evaluates audio spoofing in isolation; lacks visual cross-modal alignment

Ref.	Year	Venue & Type	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
[10]	2024	arXiv / NeurIPS (Conference)	Synthetic + pseudo-labelled real images	Depth Anything V2: zero-shot monocular depth and surface-normal estimation	State-of-the-art zero-shot depth estimation with fine detail preservation	General-purpose depth estimator; requires downstream adaptation for forensic anomaly scoring
[11]	2024	ICML (Conference)	General instruction-tuning corpora	DoRA: Directional Low-Rank Adaptation decomposing weight updates into magnitude and direction	Consistently outperforms standard LoRA on LLaMA, LLaVA, and VL-BART benchmarks	Adds a normalization step per adapted layer, marginally increasing training-time compute
[12]	2024	arXiv (Preprint)	FF++, general AIGC benchmarks	Orthogonal subspace decomposition for style–content disentanglement	Improved generalization to unseen AI-generated image styles	Requires careful multi-task loss balancing during optimization
[13]	2023	IEEE/CVF CVPR (Conference)	FF++, Celeb-DF	AltFreezing: alternating spatial and temporal feature freezing	AUC ≈ 98–99% with strong cross-manipulation generalization	Generalization drops noticeably on static single images lacking temporal motion cues
[14]	2025	arXiv (Preprint)	14 benchmark datasets (2019–2025)	GenD: parameter-efficient LayerNorm-only tuning with hyperspherical metric learning	State-of-the-art cross-benchmark generalization while updating only ≈ 0.03% of parameters	Restricted to frame-level binary classification; no spatial mask or textual explainability
[15]	2024	arXiv (Preprint)	FF++, DD-VQA	Universal forgery prompts guiding an LLM-based explanation generator	Improved BLEU/CIDEr scores for generated forensic explanations	Relies on a frozen visual backbone, reducing sensitivity to low-level sensor-noise artifacts

2.2 Derived Research Gaps

1. Semantic invariance versus low-level forensic artifact conflict: standard Vision-Language Model (VLM) backbones freeze their visual encoders to preserve semantic knowledge. This semantic invariance, however, suppresses imperceptible low-level forensic cues such as phase mismatches, sensor-noise fingerprints, and spatial blending boundaries.
2. Shortcut learning and domain overfitting: binary deepfake classifiers trained end-to-end frequently overfit to dataset-specific background attributes or compression signatures rather than to fundamental forgery characteristics, causing performance collapse on unseen generative engines.
3. Fragmentation between spatial localization and textual explainability: existing architectures operate in isolation. Specialized segmentation models (e.g., TruFor) output dense spatial masks without textual reasoning, while vision-language models generate text summaries but rely on frozen backbones that struggle with precise pixel-level localization.
4. Compute and memory bottlenecks in VLM fine-tuning: full-parameter fine-tuning of multi-billion-parameter multimodal networks requires prohibitive GPU infrastructure. Parameter-efficient fine-tuning (PEFT) methods are therefore necessary to achieve cross-domain generalization within consumer-hardware constraints (approximately 6GB of VRAM).

3. Problem Statement

Given an input facial image or video frame sequence $X = \{x_1, x_2, \ldots, x_n\}$, where each frame $x_i \in \mathbb{R}^{(H \times W \times 3)}$, together with an optional audio waveform a sampled at 16 kHz and its associated EXIF container headers, the system must segment the manipulated regions and predict media authenticity such that it concurrently outputs:

- a calibrated binary forgery-probability score, $p \in [0, 1]$;
- a dense spatial localization mask, $M \in [0, 1]^{(H \times W)}$;
- a localization-reliability confidence map, $C \in [0, 1]^{(H \times W)}$; and
- an auto-regressively generated natural-language forensic explanation, T.

These outputs must be produced under the following operational constraints: execution on a consumer workstation GPU with 8 GB of VRAM; robustness to severe social-media lossy re-compression (heavy H.264 tier); and adaptation restricted to updating no more than 5% of total backbone parameters, achieved through Directional Low-Rank Adaptation (DoRA).

Success criterion: a cross-dataset Area Under the ROC Curve, AUC ≥ 0.85 (trained on *FaceForensics++* and evaluated out-of-domain on *Celeb-DF v2* and *DFDC*), and a spatial forgery-localization mean Intersection over Union, mIoU ≥ 0.65, while demonstrating less than 10% performance degradation relative to full-parameter XceptionNet and TruFor baselines.

4. Proposed System Architecture

4.1 Pipeline Workflow Description

The end-to-end system follows a six-stage pipeline: Data Acquisition → Preprocessing → Model → Post-Processing → Evaluation → Output/Interface.

- **Data acquisition and preprocessing:** raw video frame sequences or static images are ingested. Frames are extracted via FFmpeg, facial bounding boxes are detected using RetinaFace with a fixed margin expansion, and crops are resized to a standard input resolution. In parallel, a self-supervised Noiseprint++ module filters raw crops to extract camera sensor-residual maps.
- **Dual-branch feature extraction:** RGB crops are processed through a Vision Transformer (ViT-L/14) adapted with Directional Low-Rank Adaptation (DoRA) on its linear projections to capture spatial visual anomalies. In parallel, camera noise-residual maps are passed through a four-stage convolutional encoder to capture low-level sensor fingerprints.
- **Cross-modal attention fusion:** semantic visual tokens and noise-residual embeddings are combined within a cross-modal fusion transformer using multi-head cross-attention.

- **Universal forgery prompting:** learnable Universal Forgery Prompt tokens are injected into the fused feature sequence to project spatial and spectral inconsistencies into an embedding space optimized for auto-regressive language decoding.
- **Multi-task decoding:** an MLP pooling head produces global frame-authenticity logits; transposed convolutional layers reconstruct dense pixel-localization masks and reliability maps; and a DoRA-adapted, quantized local LLM (Qwen 2.5 7B) auto-regressively generates textual forensic reports.
- **Post-processing and interface output:** sigmoid logit calibration is applied, Dempster–Shafer aggregated confidence scores are computed, spatial mask overlays are rendered, and natural-language reports are streamed to the interactive dashboard.

4.2 Pipeline Architecture Diagram

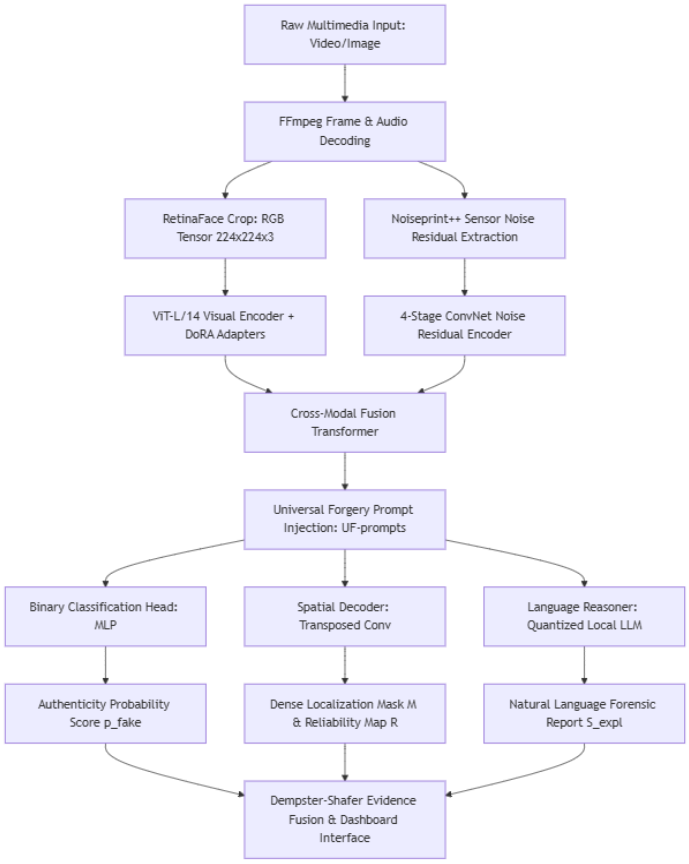


Figure 1. End-to-end PRISM pipeline architecture.

4.3 Layer-Wise Tensor Flow Specification

Module / Layer	Sub-Components & Operations	Input Shape	Output Shape	Proposed Novelty Placement
01. Preprocessor	RetinaFace detector + landmark alignment + Noiseprint++ filter	1 × 3 × H × W (raw frame)	RGB: 1×3×224×224 Noise: 1×1×224×224	Dual-stream preprocessing isolating visual attributes from sensor-residual maps.
02. Adapted Vision Encoder	ViT-L/14 patch embeddings + DoRA on QKV projections	1×3×224×224	256 × 1024 (patch tokens)	Novelty component 1: Directional Low-Rank Adaptation (DoRA) applied to ViT projections.
03. Sensor Noise Encoder	4-stage ConvNet with instance normalization and LeakyReLU	1×1×224×224	256 × 512 (noise embeddings)	Lightweight stream targeting camera sensor-residual fingerprints.
04. Cross-Modal Transformer	Multi-head cross-attention fusing flattened visual and noise embeddings	256×1024, 256×512	256 × 1024 (fused tokens)	Novelty component 2: cross-attention mechanism fusing visual features with spatial noise embeddings.
05. Universal Forgery Prompting	Learnable prompt tokens inserted via MLP projection	256×1024 + 32 prompt tokens	288 × 1024	Novelty component 3: learnable multimodal prompt module adapting unified embeddings for LLM generation.
06. Binary Classifier	LayerNorm → attention pool → Linear → Dropout → Linear	288×1024	1 (logit)	Binary classification head producing a calibrated authenticity score.
07. Dense Spatial Decoder	4 transposed-convolution blocks with skip connections	256×1024 (reshaped to 16×16×1024)	Mask: 1×224×224 Reliability: 1×224×224	Dual-output decoder predicting spatial forgery masks and confidence maps.
08. Language Reasoner	Linear projection MLP + DoRA-adapted quantized local LLM (Qwen 2.5 7B)	32 projected tokens	Variable-length text (≤256 tokens)	Auto-regressive language generation yielding natural-language forensic summaries.

4.4 Design Choice Justifications

- **Vision-side DoRA adaptation over standard LoRA:** standard Low-Rank Adaptation updates weight directions but constrains magnitude scaling, limiting flexibility when detecting subtle edge anomalies. Directional Low-Rank Adaptation decomposes weight updates into separate magnitude and directional components, improving representational flexibility at negligible additional inference cost.
- **Dedicated camera sensor-noise stream (Noiseprint++):** generative upsampling algorithms disrupt the physical Photo-Response Non-Uniformity (PRNU) noise patterns inherent to digital camera sensors. Extracting self-supervised camera noise residuals ensures that spatial forgery localization remains robust under lossy H.264 compression.
- **Universal forgery prompting:** passing raw visual tokens directly to a language decoder often causes modality misalignment. Learnable prompt tokens project multimodal features directly into the text decoder's embedding space, improving natural-language explanation quality.

- **Dempster–Shafer evidence combination with quality discounting:** standard score averaging fails when agent evaluations conflict due to environmental noise. Discounting belief masses using visual- and acoustic-quality indices reallocates unassigned belief mass to an explicit uncertainty term, preventing false-positive detections on degraded inputs.

4.5 Novelty-Component Architecture Diagram

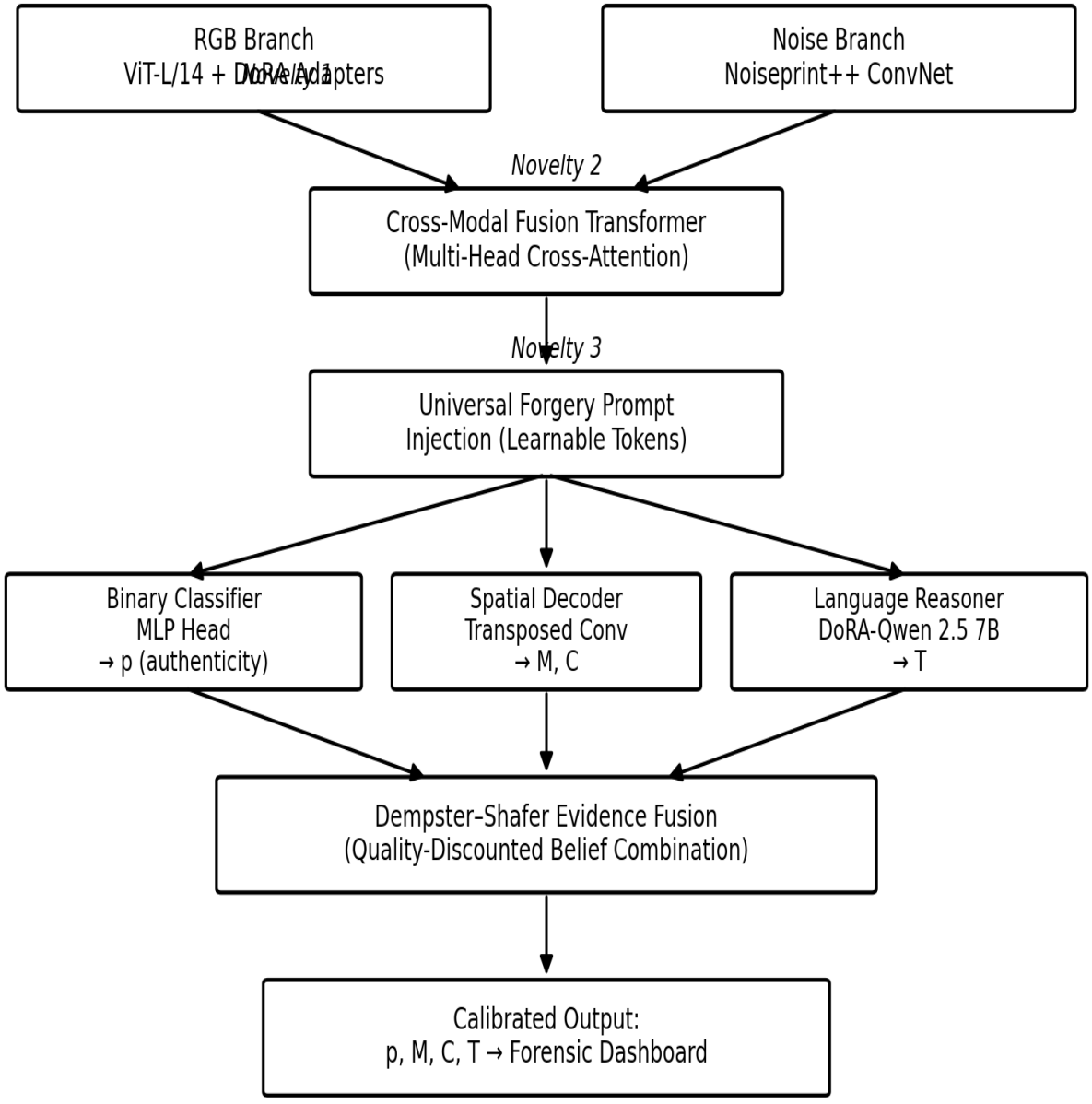


Figure 2. Internal model architecture highlighting the three proposed novelty components.

5. Planned Experimental Setup and Feasibility Note

5.1 Experimental Setup

- **Benchmark datasets:** *FaceForensics++* (in-domain training and validation); *Celeb-DF v2*, *DFDC*, and *WildDeepfake* (out-of-domain cross-dataset testing).
- **Data split strategy:** a subject-isolated 80% training, 10% validation, and 10% testing partition, designed to prevent identity leakage across identical source footage.
- **Evaluation metrics:** classification performance via AUC and Equal Error Rate (EER); localization performance via mean Intersection over Union (mIoU) and pixel-level F1; and text quality via BLEU-4, CIDEr, and ROUGE-L evaluated against reference explanations.
- **Baselines for comparison:** XceptionNet, TruFor, AltFreezing, and two recent LLM-explanation and parameter-efficient baselines drawn from the literature survey.

5.1.1 Experimental Evaluation Workflow

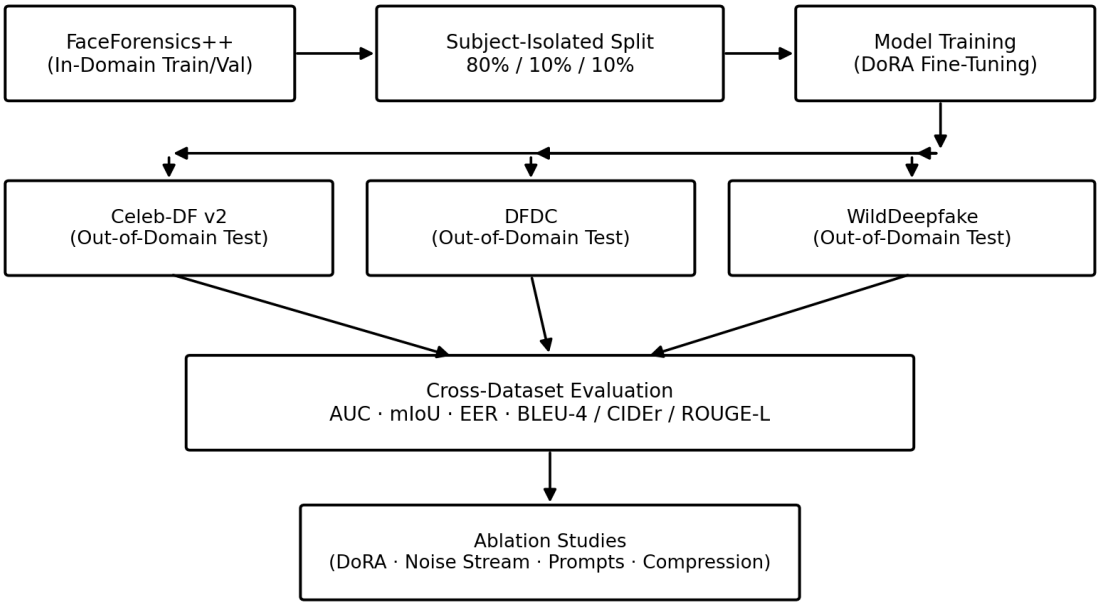


Figure 3. Cross-dataset training, testing, and ablation evaluation workflow.

Planned Ablation Studies

- Impact of vision-side DoRA adaptation: evaluating detection AUC and localization mIoU across frozen encoders, standard LoRA, and DoRA tuning.
- Ablation of the sensor-noise stream: removing the Noiseprint++ encoder to quantify performance drops on compressed media.
- Contribution of Universal Forgery Prompts: measuring LLM text-explanation scores with and without learnable prompt tokens.

- Robustness to image degradation: evaluating performance decay across progressive H.264 compression tiers (raw, light, heavy).

5.2 Feasibility Note

Hardware Availability and Memory Optimization

- Local workstation: Intel Core i7 processor, 16 GB DDR5 RAM, NVIDIA GeForce RTX GPU with 8 GB GDDR6 VRAM.
- Cloud acceleration: free-tier cloud environments (Kaggle Notebooks) offering roughly 30 hours per week of dedicated NVIDIA Tesla P100 / T4 GPUs (16 GB VRAM) for large-scale training runs.
- VRAM memory management: dynamic process-level model offloading (periodic cache clearing and garbage collection) keeps peak memory consumption below approximately 6.5 GB VRAM, ensuring execution stability on 8 GB GPUs.

Dataset Storage and Preprocessing Strategy

To resolve local storage limitations and mitigate disk I/O bottlenecks during training:

- Video files are processed offline in CPU mode.
- Frames are uniformly sampled at 32 frames per video, cropped to aligned facial regions, and cached locally in WebP format.
- This preprocessing pipeline compresses the dataset storage footprint from over 500 GB of raw video to approximately 35 GB of cached tensors.

Estimated Training Time

By restricting parameter updates to DoRA adapter weights, representing under 5% of total network parameters, and utilizing FP16 automatic mixed precision, the model is expected to require approximately 2 to 3 GPU-hours per epoch on a 16 GB GPU, with convergence reached within 15 training epochs (roughly 30 to 45 total GPU-hours), fitting comfortably within standard weekly cloud compute allocations.

Risk Management Matrix

Identified Technical Risk	Severity	Concrete Fallback and Mitigation Strategy
CUDA out-of-memory (OOM) errors	High	Enable PyTorch gradient checkpointing (model.gradient_checkpointing_enable()); reduce the micro-batch size to 4 with 4 gradient-accumulation steps; employ 4-bit quantized local LLMs via Ollama (ollama.com).
Shortcut learning on background artifacts	Medium	Enforce paired real–fake training in which manipulated samples are strictly paired with their authentic source-video counterparts; apply background-masking data augmentation.
Cloud GPU session timeouts	Low	Implement automated state checkpointing every 5 epochs; save checkpoints directly to external cloud storage to preserve training state across session restarts.

6. Conclusion and Next Steps

This report has established the motivation, technical landscape, and formal problem definition for PRISM, a cross-domain multimodal deepfake detection and forensic reasoning system. The literature survey identifies a consistent fragmentation in existing work between accurate spatial localization, robust cross-dataset generalization, and human-readable forensic explanation, a gap that directly motivates the proposed dual-branch, prompt-mediated architecture.

The next phase of this project (DA2) will focus on implementing the preprocessing and dual-branch feature-extraction stages, establishing baseline reproductions of XceptionNet and TruFor on FaceForensics++, and validating the feasibility assumptions described in Section 5 on the available local and cloud hardware before proceeding to full multi-task training of the fusion and language-reasoning components.

7. Contribution Matrix

Member	Focus Area	Research & Literature Contribution
Tanush Bhootra (24BRS1282)	System Architecture & Model Fusion	• Literature analysis on PEFT fine-tuning (DoRA paper) • Research on Cross-Modal Fusion mechanisms
Somil Jain (24BRS1346)	Vision Stream & Spatial Masking	• Survey of deepfake benchmarks (FF++, Celeb-DF, DFDC) • Research on spatial localization models (TruFor)
Kulsum Malik (24BRS1319)	LLM Reasoning & User Interface	• Research on explainable deepfake reporting (FFAA paper) • Survey of audio anti-spoofing techniques (AASIST)



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